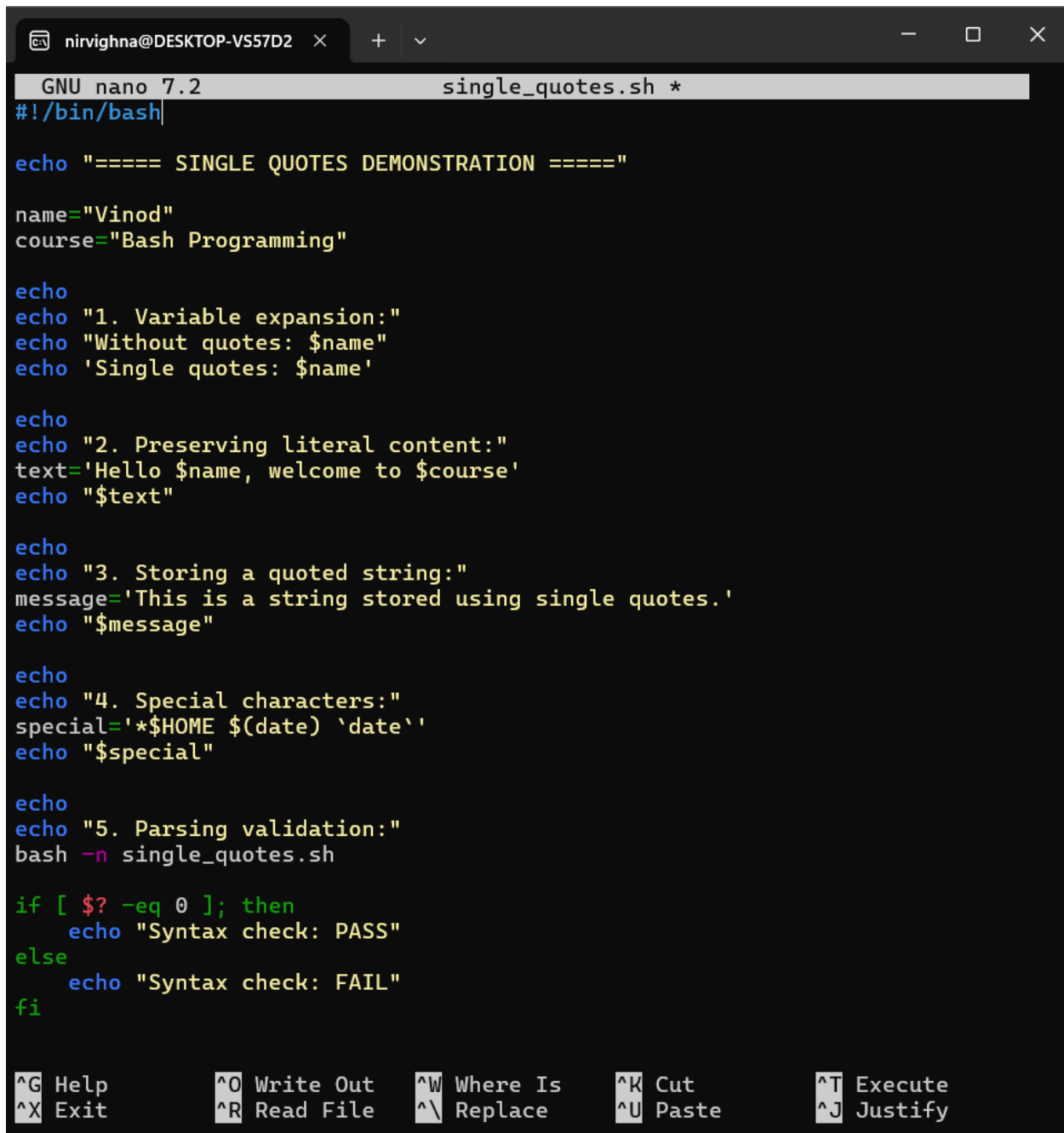


SKILL -5:

To Apply single Quotes, Preserve Literal Content ignore variable expansion store quoted strings validate parsing result test edges cases.



```
GNU nano 7.2 single_quotes.sh *
#!/bin/bash

echo "===== SINGLE QUOTES DEMONSTRATION ====="

name="Vinod"
course="Bash Programming"

echo
echo "1. Variable expansion:"
echo "Without quotes: $name"
echo 'Single quotes: $name'

echo
echo "2. Preserving literal content:"
text='Hello $name, welcome to $course'
echo "$text"

echo
echo "3. Storing a quoted string:"
message='This is a string stored using single quotes.'
echo "$message"

echo
echo "4. Special characters:"
special='*$HOME $(date) `date`'
echo "$special"

echo
echo "5. Parsing validation:"
bash -n single_quotes.sh

if [ $? -eq 0 ]; then
    echo "Syntax check: PASS"
else
    echo "Syntax check: FAIL"
fi

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
nirvighna@DESKTOP-VS57D2  ×  +  ▾  -  □  ×  
nirvighna@DESKTOP-VS57D2I:~$ mkdir -p ~/Session5  
cd ~/Session5  
nirvighna@DESKTOP-VS57D2I:~/Session5$ PWD  
PWD: command not found  
nirvighna@DESKTOP-VS57D2I:~/Session5$ ped  
Command 'ped' not found, but there are 16 similar ones.  
nirvighna@DESKTOP-VS57D2I:~/Session5$ pwd  
/home/nirvighna/Session5  
nirvighna@DESKTOP-VS57D2I:~/Session5$ nano single_quotes.sh  
nirvighna@DESKTOP-VS57D2I:~/Session5$ chmod +x single_quotes.sh  
nirvighna@DESKTOP-VS57D2I:~/Session5$ ./single_quotes.sh  
===== SINGLE QUOTES DEMONSTRATION =====  
  
1. Variable expansion:  
Without quotes: Vinod  
Single quotes: $name  
  
2. Preserving literal content:  
Hello $name, welcome to $course  
  
3. Storing a quoted string:  
This is a string stored using single quotes.  
  
4. Special characters:  
*$HOME $(date) `date`  
  
5. Parsing validation:  
Syntax check: PASS  
  
6. Edge case:  
Single quotes preserve $variables and command substitution literally.  
  
===== END =====  
nirvighna@DESKTOP-VS57D2I:~/Session5$ |
```

```
nirvighna@DESKTOP-VS57D2  ×  +  v
GNU nano 7.2 double_quotes.sh *
#!/bin/bash

echo "===== DOUBLE QUOTES DEMONSTRATION ====="

name="Vinod Kumar"
course="Shell Scripting"

echo
echo "1. Variable expansion:"
echo "Name: $name"
echo "Course: $course"

echo
echo "2. Preserving spaces:"
message="Welcome to Bash Shell Programming"
echo "$message"

echo
echo "3. Command substitution:"
current_date="$(date)"
echo "Current date and time: $current_date"

echo
echo "4. Nested tokens:"
filename="student report.txt"
echo "Filename: $filename"

echo
echo "5. Quoted command:"
command="echo Hello from Bash"
echo "Stored command: $command"
$command

echo
echo "6. Parsing validation:"
bash -n double_quotes.sh

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

```
nirvighna@DESKTOP-VS57D2I:~/Session5$ ./double_quotes.sh
===== DOUBLE QUOTES DEMONSTRATION =====

1. Variable expansion:
Name: Vinod Kumar
Course: Shell Scripting

2. Preserving spaces:
Welcome to Bash Shell Programming

3. Command substitution:
Current date and time: Sat Sep 26 06:15:06 UTC 2026

4. Nested tokens:
Filename: student report.txt

5. Quoted command:
Stored command: echo Hello from Bash
Hello from Bash

6. Parsing validation:
Syntax check: PASS

7. Output validation:
Name validation: PASS

8. Edge case:
The value contains spaces and $variable remains literal.

===== END =====
```

SKILL 6:

To create process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

```
nirvighna@DESKTOP-VS57D2  x  +  v  -  □  ×

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 64

void parse_input(char *input, char *args[])
{
    int argc = 0;
    char *src = input;
    char *dst = input;

    while (*src != '\0' && argc < MAX_ARGS - 1)
    {
        while (*src == ' ' || *src == '\t' || *src == '\n')
            src++;

        if (*src == '\0')
            break;

        args[argc++] = dst;

        int escaped = 0;
        while (*src != '\0')
        {
            if (escaped)
            {
                *dst++ = *src++;
                escaped = 0;
            }
            else if (*src == '\\')
            {
                escaped = 1;
                src++;
            }
        }
    }

    EOF return 0; session 6 shell terminated.\n"); atus: %d\n", ;') '\n')
```

```

    }
EOF return 0;session 6 shell terminated.\n");atus: %d\n",;') '\n')
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l src/session6_shell.c
-rw-r--r-- 1 nirvighna nirvighna 2562 Sep 27 12:33 src/session6_shell.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |

```

```

nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l bin/session6_shell
-rwxr-xr-x 1 nirvighna nirvighna 20560 Sep 27 12:34 bin/session6_shell
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/session6_shell
Session 6 Mini Shell
Type 'exit' to quit.
session6-shell> |

```

```

nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/session6_shell
Session 6 Mini Shell
Type 'exit' to quit.
session6-shell> ls
Parsed arguments:
argv[0] = [ls]
Parent process PID: 3463
Created child PID: 3468
Child process PID: 3468


|           |                |                     |                   |          |
|-----------|----------------|---------------------|-------------------|----------|
| Makefile  | mini_shell.c   | producer_consumer   | strace_output.txt | zombie   |
| bin       | nano.2750.save | producer_consumer.c | wait_demo         | zombie.c |
| input.txt | output.txt     | src                 | wait_demo.c       |          |


Child exited with status: 0
session6-shell> |

```

```

Child exited with status: 0
session6-shell> echo One\ Two Three\ Four Five
Parsed arguments:
argv[0] = [echo]
Parent process PID: 3463
Created child PID: 3476
Child process PID: 3476

Child exited with status: 0
session6-shell> ls -l
Parsed arguments:
argv[0] = [ls]
Parent process PID: 3463
Created child PID: 3477
Child process PID: 3477


|           |                |                     |                   |          |
|-----------|----------------|---------------------|-------------------|----------|
| Makefile  | mini_shell.c   | producer_consumer   | strace_output.txt | zombie   |
| bin       | nano.2750.save | producer_consumer.c | wait_demo         | zombie.c |
| input.txt | output.txt     | src                 | wait_demo.c       |          |


Child exited with status: 0
session6-shell> ^[[2~|

```